

ORF307_HW6_soln

ORF307 Homework 6

Due: Friday, November 7, 2025 11:59 pm ET

- Please export your code with output as pdf.
- If there is any additional answers, please combine them as **ONE** pdf file before submitting to the Gradescope.

Question 1

Let A be a given matrix. Show that exactly one of the following alternatives must hold.

- (a) There exists some $x \neq 0$ such that $Ax = 0, x \geq 0$.
- (b) There exists some y such that $A^T y > 0$.

Question 1 Solution

Construct the following linear program:

$$\begin{aligned} & \text{minimize} && -\mathbf{1}^T x \\ & \text{subject to} && Ax = 0 \\ & && x \geq 0. \end{aligned} \tag{1}$$

Then problem (1) has dual

$$\begin{aligned} & \text{maximize} && 0^T y \\ & \text{subject to} && A^T y \geq \mathbf{1}. \end{aligned} \tag{2}$$

In the primal, $x = 0$ is feasible, so strong duality holds since $p^* \neq +\infty$. Then, there are two alternatives.

If there exists y that satisfies (b), then the optimal solution of (2) is 0. If some y satisfies $A^T y > 0$, then we can define $z = y / \min_i |(A^T y)_i|$ such that $A^T z \geq \mathbf{1}$. This means that (b) corresponds to (2) being feasible. By strong duality, (1) also has optimal solution 0. But since $-\mathbf{1} < 0$, then $c^T x = 0$ if and only if $x = 0$, which violates (a).

If there exists no y that satisfies (b), then the optimal solution of (2) is $-\infty$, i.e. the dual is infeasible. This can only happen if the primal is unbounded. The primal is unbounded if there exists some $x \neq 0$ such that $Ax \geq 0, x \geq 0$, because then $z = \alpha x$ for $\alpha > 0$ is also a solution. This shows that condition (a) holds.

Lastly, to show that the alternatives cannot be true together, if we assume such an x, y both exist (that is, $x \neq 0, Ax = 0, x \geq 0, A^T y > 0$), then

$$0 = y^T(Ax) = (A^T y)^T x > 0,$$

which is a contradiction.

Question 2

An alternative to the phase-I/phase-II method for solving the LP

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax = b, \\ & && x \geq 0 \end{aligned} \tag{3}$$

is the “big-M”-method, in which we solve the auxiliary problem

$$\begin{aligned} & \text{minimize} && c^T x + M\mathbf{1}^T z \\ & \text{subject to} && Ax + z = b \\ & && x \geq 0, z \geq 0, \end{aligned} \tag{4}$$

where $M > 0$ is a parameter and $z \in \mathbf{R}^m$ is an auxiliary variable. Here, $A \in \mathbf{R}^{m \times n}, c \in \mathbf{R}^n$, and $b \in \mathbf{R}^m$. Note that this auxiliary problem has an initial basic feasible solution $(x, z) = (0, b) \geq 0$.

(a) Derive the dual LP of (4).

(b) Prove the following property:

If $M > -y_i^*$ for $i = 1, \dots, m$, where y^* is an optimal solution of the dual of (3), then the optimal z in (4) is zero, and therefore the optimal x in (4) is also an optimal solution of (3).

Hint: Use complementary slackness.

Q2 Solution

(a) The dual to (4) can be written as

$$\begin{aligned} & \text{maximize} && -b^T y \\ & \text{subject to} && A^T y + c \geq 0 \\ & && Iy + M\mathbf{1} \geq 0 \end{aligned} \tag{5}$$

(b) Recognize that the constraint $Iy + M\mathbf{1} \geq 0$ corresponds to the primal variable z . So, by complementary slackness, for each i , for an optimal y^* and z^* , it must be that

$$z_i^*(y_i^* + M) = 0$$

By assumption, $M > -y_i^*$, so $y_i^* + M \neq 0$, so it must be that $z_i^* = 0$. This must hold for each i . Therefore, the optimal x in (4) is also an optimal solution of (3).

Another valid solution uses strong duality. At optimality, we have zero duality gap, which means:

$$\begin{aligned}
c^T x + b^T y &= \begin{bmatrix} c \\ M\mathbf{1} \end{bmatrix}^T \begin{bmatrix} x \\ z \end{bmatrix} + \left(\begin{bmatrix} A & I \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} \right)^T y \\
&= \begin{bmatrix} c \\ M\mathbf{1} \end{bmatrix}^T \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} x \\ z \end{bmatrix}^T \begin{bmatrix} A^T y \\ Iy \end{bmatrix} \\
&= \begin{bmatrix} x \\ z \end{bmatrix}^T \begin{bmatrix} c - A^T y \\ M\mathbf{1} - Iy \end{bmatrix} \\
&= 0.
\end{aligned}$$

Since $M > y_i^*$ for $i = 1, \dots, m$, $M\mathbf{1} - Iy > 0$, which implies $z = 0$, therefore the optimal x in (4) is also an optimal solution of (3).

Question 3

Consider the following LP:

$$\begin{aligned}
&\text{minimize} && 13x_1 + 10x_2 + 6x_3 \\
&\text{subject to} && 5x_1 + x_2 + 3x_3 = 8 \\
&&& 3x_1 + x_2 = 3 \\
&&& x_1, x_2, x_3 \geq 0.
\end{aligned} \tag{7}$$

- Solve it using the big-M formulation as in Q3 obtaining optimal primal and dual variables (use the provided function).
- Derive the dual LP for (7)
- Solve the dual using CVXPY and compare the optimal primal-dual variables with the ones from (a).

```
[9]: '''
This code is provided to help with question 4.
This code returns optimal primal variables x
and dual variables y.
'''
import cvxpy as cp
import numpy as np
import numpy.linalg as la

def simplex_iteration(x, B, problem):
    """Perform one simplex iteration given
    - basic feasible solution x
    - basis B

    It returns new x, new basis, new dual variable,
    and termination flag (true/false)
    """
    A, b, c = problem['A'], problem['b'], problem['c']
    m, n = A.shape
    A_B = A[:, B]
```

```

# Compute reduced cost vector
p = la.solve(A_B.T, c[B])
c_bar = c - A.T @ p

# Check optimality
if np.all(c_bar >= 0):
    print("Optimal solution found!")
    return x, B, -p, True

# Choose j such that c_bar < 0 (first one)
j = np.where(c_bar < 0)[0][0]

# Compute search direction d
d = np.zeros(n)
d[j] = 1
d[B] = la.solve(A_B, -A[:, j])

# Check for unboundedness
if np.all(d >= 0):
    print("Unbounded problem!")
    return None, None, True

# Compute step length theta
d_i = np.where(d[B] < 0)[0]
theta = np.min(- x[B[d_i]] / d[B[d_i]])
i = B[d_i[np.argmin(- x[B[d_i]] / d[B[d_i]])]]

# Compute next point
x_next = x + theta * d

# Compute next basis
B_next = B
B_next[np.where(B == i)[0]] = j

return x_next, B_next, -p, False

def simplex_algorithm(x, B, problem, max_iter=1000):
    """Run simplex algorithm"""

    for k in range(max_iter):

        x, B, y, end = simplex_iteration(x, B, problem)

        if end:
            break
    return x, B, y

```

Q3 Solution

(a)

```
[10]: # Original problem
c = np.array([13, 10, 6])
A = np.array([[5, 1, 3],
              [3, 1, 0]])
b = np.array([8, 3])

# Transformed problem
M = 100
c_tilde = np.append(c, M * np.ones(2))
A_tilde = np.hstack([A, np.eye(2)])
b_tilde = b

x0 = np.append(np.zeros(3), b)
B0 = np.array([3, 4])

[11]: x_sol, B_sol, y_sol = simplex_algorithm(x0, B0, {'A': A_tilde, 'b': b_tilde,
↪ 'c': c_tilde})
```

Optimal solution found!

```
[12]: y_sol
```

```
[12]: array([-2., -1.])
```

(b) The dual is

$$\begin{aligned} &\text{maximize} && -8y_1 - 3y_2 \\ &\text{subject to} && 5y_1 + 3y_2 \geq -13 \\ & && y_1 + y_2 \geq -10 \\ & && 3y_1 \geq -6 \end{aligned}$$

```
[13]: # Solve using CVXPY
import cvxpy as cp
y = cp.Variable(2)
objective = cp.Maximize(-b @ y)
constraints = [A.T @ y + c >= 0]
problem = cp.Problem(objective, constraints)
problem.solve(verbose=True)
```

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It	pcost	dcost	gap	pres	dres	k/t	mu	step	sigma
IR	BT								
0	-2.635e+01	-2.659e+01	+2e+01	3e-01	5e-03	1e+00	6e+00	---	---
1	1 - - -								

```

1  -1.928e+01  -1.909e+01  +4e+00  5e-02  7e-04  4e-01  1e+00  0.8791  6e-02
1  1  1 | 0  0
2  -1.900e+01  -1.900e+01  +4e-02  5e-04  8e-06  4e-03  1e-02  0.9890  1e-04
1  1  1 | 0  0
3  -1.900e+01  -1.900e+01  +4e-04  6e-06  9e-08  5e-05  1e-04  0.9890  1e-04
1  0  0 | 0  0
4  -1.900e+01  -1.900e+01  +5e-06  7e-08  1e-09  5e-07  1e-06  0.9890  1e-04
1  0  0 | 0  0
5  -1.900e+01  -1.900e+01  +5e-08  7e-10  1e-11  6e-09  2e-08  0.9890  1e-04
1  0  0 | 0  0

```

OPTIMAL (within feastol=7.3e-10, reltol=2.8e-09, abstol=5.3e-08).
Runtime: 0.000251 seconds.

[13]: 19.000000004942304

[14]: y.value

[14]: array([-2. , -1.00000001])